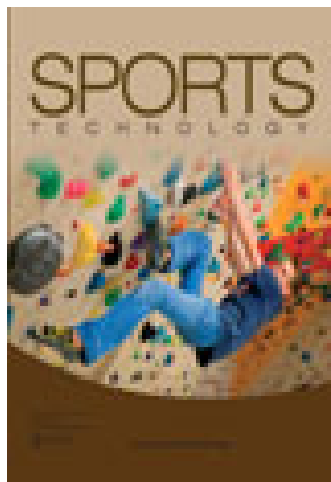




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RESEARCH ARTICLE

## The effects of two maximum grip strength training methods using the same effort duration and different edge depth on grip endurance in elite climbers

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### Abstract

Nine experienced rock climbers (mean climbing ability of 8a + /b) were randomly assigned to Group A ( $n = 5$ ) and Group B ( $n = 4$ ). Both groups trained dead hanging using two different methods. One method consisted of using the minimum edge depth (MED) they could hold the weight of their body; the other consisted of using a bigger edge (18 mm) with maximum added weight (MAW). Group A performed MED from Weeks 1 to 4, and then performed MAW the following 4 weeks (termed as MED–MAW group); Group B performed MAW from Weeks 1 to 4 and then performed MED the following 4 weeks (termed as MAW–MED group). Maximum grip strength and endurance tests were conducted initially (ST1; ET1), following 4 weeks (ST2; ET2), 8 weeks (ST3; ET3), 2 weeks (ST4; ET4) and 4 weeks (ST5; ET5) completion of training to determine the effects of detraining. The 9.6% improvement in grip strength ( $p > 0.05$ ) in MAW–MED group in ST2 and 6.9% in ST4 was greater than in MED–MAW group. In terms of grip endurance, MAW–MED group in ET2 (16.69%) and in ET3 (19.95%) improved more than MED–MAW group ( $p > 0.05$ ). Significant positive correlation was found between ST and ET, and between changes in strength and changes in endurance at all stages, controlling for body weight in all cases. The present data suggest that the most effective sequence of finger strength training methods is MAW–MED.

**Keywords:** rock climbing, finger strength training, finger endurance training, elite climbers, strength on endurance

### Introduction

There are a number of studies that attempt to describe the elite climber in discovering the determining factors of performance in the sport of rock climbing. Some authors suggest that the physical limits of the rock climber are those of maximum finger strength (Grant, Hynes, Whittaker & Aitchison, 1996; Guidi, 1994; Lehner & Heyters, 1998; MacLeod, Sutherland, Buntin, Whitaker, Aitchison, Bradley & Grant, 2007; Watts & Jensen, 2003) and finger endurance (Cutts & Bollen, 1993; Ferguson & Brown, 1997; Grant, Shields, Fitzpatrick, Ming Loh, Whitaker, Watt & Kay, 2003; Usaj, 1996, 2001). It has also been shown that maximum strength training provokes improvement in muscular endurance (Hennessy, Watson, Orlando, Florida, Hickson, Dvorak, Gorostiaga, Kurowski, & Foster, 1988; Hickson, Dvorak, Gorostiaga, Kurowski, Foster, Hidaka & Foster (1988, 1994); Marcinik, Potts, Schlabach, Will, Dawson & Hurley, 1991; Osteras, Helgerud & Hoff, 2002; Paavolainen, Hakkinen, Hamalainen, Nummela & Rusko, 1999; Stone & Coulter, 1994). Furthermore, a

correlation was found between the maximum voluntary contraction (MVC) of the hand and the time until fatigue in a second contraction at 60% of MVC after an incomplete pause, according to Guidi (1994), as well as between the MVC using the open crimp and the performance level in the study by MacLeod et al. (2007). All the above findings lead us to suggest that training and increasing maximum grip strength can be important for the improvement in maximum grip endurance in rock climbing and also for an increase in performance.

It has also been suggested that to produce this effect, it is necessary for the strength training to be specific (Anderson & Kearney, 1982; Bell, Snyder, Neary & Quinney, 1989; Paavolainen et al. 1999; Tanaka, Costill, Thomas, Fink & Widrick, 1993). We observe that in the majority of studies, a dynamometer has been used to measure or study hand endurance (Binney, 2002; Usaj, 2001; Quaine & Vigouroux, 2004; Quaine, Vigouroux & Martin, 2003; Watts & Jensen, 2003), which is questionable from a performance point of view according to some authors (Schöffl, Möckel, Köstermeyer, Roloff & Küpper, 2005; Schweizer, Schneider

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& Goehner, 2007; Watts, 2004) because applying force on the dynamometer causes the second phalange of the fingers to flex against the palm of the hand, a gesture which seldom occurs in climbing. Therefore, as also other authors suggest (Grant et al. 1996; MacLeod et al. 2007; Schweizer et al., 2007; Watts, Newbury & Sulentic, 1996; Jensen, Watts, Lawrence, Moss & Wagonsomer, 2005), it would be necessary to propose the use of a specific device that could measure – and be used for – finger strength and endurance training a specific rock climbing movement technique.

An essential characteristic of climbing is hanging off small edges or pockets, and at a high level we consider a small grip edge to be of a depth less than half the size of the distal phalange of the fingers (Quaine & Vigouroux, 2004; Schweizer & Hudek, 2011; Watts, 2004). The usual way of gripping such a hold is the so-called crimp grip, where the distal interphalangeal is hyper-extended and proximal interphalangeal joints are rather 80–90° flexion in half crimp position, against > 90° in full crimp (Schweizer, 2001). These observations support the suggestion that in order to achieve better climbing performance a grip strength method can be used, allowing hanging off holds with the least or minimum edge depth (MED) as possible. On the other hand, regarding improvements in strength, it has been observed in a number of studies that training with extra weight (maximum added weight, MAW) improves strength (Behm, 1995; Hakkinen, Alen & Komi, 1985b; Hakkinen, Pakarinen, Kraemer, Hakkinen, Valkeinen & Alen, 2001; Harris, Stone, O'Bryant, Prolux & Johnson, 2000; Kraemer, Koziris, Ratamess, Hakkinen, Triplett-McBride, Fry & Fleck, 2002; Siff & Verkoshansky, 1999). However, only a few scientific articles address the topic of specific training to optimise the rock climber's performance (Koestermeyer & Weineck, 1995; Schweizer et al. 2007), and it appears that none of them have attempted to compare the effectiveness of the two specific maximum finger strength training regimes mentioned: using MED or MAW in high-level climbers. Additional research is needed to assess how specific climbing training impacts on a climber's performance (Watts, 2004).

Therefore, the aim of this research was to determine the effects of two different strength training methods in which either the edge depth or the added weight is variable, carried out in a different order, over grip endurance on an exercise specific of climbing, on a group of elite climbers.

## Methods

Various prerequisites were established for the participants to take part in the study (Table I). Each participant was individually informed in writing about the characteristics, risks and drawbacks associated with the study and was given a consent

Table I. Requirements for the participants in the study.

- |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• To be at least 25 years of age</li> <li>• To have been climbing for at least 5 years</li> <li>• Regular training practice for climbing, more than two days a week during the past year</li> <li>• Rock climbing level of 8a or more</li> <li>• Experience with dead-hanging training on edges for more than 4 weeks</li> <li>• Not having performed dead-hanging training in the previous 2 months</li> <li>• Absence of finger or lower back injuries during the past year, or any condition that makes it inadvisable to take part in a training program</li> <li>• Being able to commit themselves to a training and testing schedule for 3 months</li> </ul> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

form to sign. Participants who did not meet any of the requirements were excluded.

The voluntary participants completed a questionnaire detailing their personal data, climbing grade performance, defined as the most difficult redpoint ascent achieved in the past 6 months ("redpoint" means leading a sport route after inspecting it, and often after practising individual moves), number of years of training for climbing and age. Climbing ability was converted to a standard numerical scale to enable calculations and statistical analysis according to Watts, Martin & Durtschi (1993) (Table II).

After the selection, 12 sport climbers (10 men and 2 women) voluntarily joined the study; 9 of them (8 men and 1 woman) finished the programme. The descriptive figures for each group are shown in Table III.

## Apparatus

We designed and fabricated a specific apparatus that consists of an edge made of wood whose depth can be

Table II. Standardized climbing ability conversion chart.

| French scale | Standard numerical scale |
|--------------|--------------------------|
| 6a           | 1                        |
| 6a+          | 1.25                     |
| 6b           | 1.50                     |
| 6b+          | 1.75                     |
| 6c           | 2                        |
| 6c+          | 2.25                     |
| 7a           | 2.50                     |
| 7a+          | 2.75                     |
| 7b           | 3                        |
| 7b+          | 3.25                     |
| 7c           | 3.50                     |
| 7c+          | 3.75                     |
| 8a           | 4                        |
| 8a+          | 4.25                     |
| 8b           | 4.50                     |
| 8b+          | 4.75                     |
| 8c           | 5                        |
| 8c+          | 5.25                     |
| 9a           | 5.50                     |
| 9a+          | 5.75                     |
| 9b           | 6                        |

Note: Reproduced from Watts et al. (1993).

Table III. Descriptive characteristics of the participants.

|                                                                           | Group A<br>(n = 5)<br>Mean ± sd | Group B<br>(n = 4)<br>Mean ± sd |
|---------------------------------------------------------------------------|---------------------------------|---------------------------------|
| Age (years)                                                               | 32.20 ± 4.32                    | 28.25 ± 2.06                    |
| Height (cm)                                                               | 171.40 ± 10.91                  | 168.88 ± 4.48                   |
| Initial body mass (kg)                                                    | 63.68 ± 11.98                   | 68.95 ± 4.87                    |
| Climbing experience (years)                                               | 15.80 ± 4.60                    | 11.25 ± 2.22                    |
| Climbing grade performance<br>(standardized scale;<br>Watts et al., 1993) | 4.30 ± 0.57                     | 4.19 ± 0.75                     |

Climbing grade performance = the most difficult redpoint ascent achieved in the past 6 months (“redpoint” means leading a sport route after inspecting it, and often after practising individual moves).

adjusted with a precision gauge to measure finger endurance and strength, and designed to train in a specific way (dimensions 500 × 250 × 24 mm, manufactured by Eva López & Dafnis Fernández in 2004; Creative Commons license; Figure 1). In addition, two Casio® hand chronometers were used to measure the hanging duration, a Petzl® harness to hang the added weights and magnesium carbonate of the brand Camp® to reduce the effect of perspiration.

Design

During the first week (week 1) initial maximum finger strength (ST) and finger endurance tests (ET) were carried out. The tests were repeated (retests) in the second week (week 2) to assess the reliability of the measurements and served as initial tests (ST1 and ET1). The participants were sorted according to their results in the retest and then assigned to one of the two groups using the *abba* sorting: a participant was randomly chosen and assigned to Group A and



Figure 1. Specific apparatus consisting of an adjustable edge depth. The depth of the edge can be changed with the help of the screws and measured with a calliper. (Dimensions 500 × 250 × 24 mm; manufactured by Eva López & Dafnis Fernández in 2004; Creative Commons license).

Table IV. Training schedule.

|         |                       |                                   |
|---------|-----------------------|-----------------------------------|
| Week 1  | ST and ET (tests)     | Pre-test phase (test and retest)  |
| Week 2  | ST1 and ET1 (retests) | Training phase 1                  |
| Week 1  | 3 × 10”:3'            | A group: MED                      |
| Week 2  | 4 × 10”:3'            | B group: MAW                      |
| Week 3  | 5 × 10”:3'            |                                   |
| Week 4  | 5 × 10”:3'            |                                   |
| Week 5  | ST2 and ET2           | Test post-training phase 1        |
| Week 6  | 3 × 10”:3'            | Training phase 2                  |
| Week 7  | 4 × 10”:3'            | Group A:MAW                       |
| Week 8  | 5 × 10”:3'            | Group B:MED                       |
| Week 9  | 5 × 10”:3'            |                                   |
| Week 10 | ST3 and ET3           | Test post-training phase 2        |
| Week 11 |                       | Rest from MAW and MED: detraining |
| Week 12 | ST4 and ET4           |                                   |
| Week 13 |                       |                                   |
| Week 14 | ST5 and ET5           |                                   |

ST = strength test; ET = endurance test; MED = training with the minimum edge depth without added weight; MAW = Training with maximum added weight on 18 mm edge.

the rest were assigned according to the *bba* pattern until both groups were complete.

In Week 5, tests for maximum strength and endurance (ST2 and ET2) were carried out and in Week 10, ST3 and ET3 tests were carried out. To assess the effects of detraining and after a rest week from the training methods (week 11), ST4 and ET4 tests were performed in Week 12. The following week the participants rested, and a week later (week 14) the final tests were carried out (ST5 and ET5; Table IV).

Testing procedure

Before measurements, participants were informed about the procedures and rules for the tests and were instructed regarding the proper hanging technique in the 2 weeks before the first test. Two days before the test each participant performed a light training session, having rested from training the day before the test. Tests were carried out on Tuesdays, always at the same time of day.

Before the test itself, a specific 15-min warm-up was performed, consisting of neck, shoulder, arm and finger flexing and extensions, and three progressive sets of the test exercise with moderate intensity. The participants were weighed and their heights were recorded after a 5-min rest period. Then the test for maximum strength with added weight was carried out, and after a 10-min rest the endurance test was performed.

During the weeks when the participants rested from the proposed training in this study, they continued with their individual training that was similar, at an intensity specific to their performance level.

*Strength test (ST)*. It consisted of half crimp grip dead hanging with an edge depth of 15 mm and the

maximum load with which the participant could maintain the grip for 5 s (Figure 2). The 15 mm edge depth (Figure 3) was considered to be the most representative of the average size of holds used in competitions, and to have a high significant relationship with climbing grade. With this size as a basis, a bigger depth (18 mm) was chosen for the weekly training and a smaller one (11 mm; Figure 4) for the endurance test, given that according to Bourne, Halaki, Vanwanseele & Clarke, (2011), holding force for very small holds (smaller than 5 mm) depends on finger anatomy (soft tissue) rather than on muscular strength.

After a warm-up, the strength test was performed. An initial added weight was chosen so that the participant could hold onto the edge for 15–20 s. Progressive attempts were made with weight increments of 5–10 kg according to the participants' capacity with rest intervals of 5 min. The idea of the increments was to keep the number of attempts below five in order to reduce fatigue.

When the participant was no longer able to maintain the half crimp for 5 s with all their fingers, either flexing, lifting their arms or trunk or legs up, the test was stopped and the last load withstood was recorded.

*Endurance test (ET).* It consisted of half crimp dead hanging on an edge with a depth of 11 mm for the maximum possible duration without additional weight (Figure 5). The test ended once the contact was lost between the participant and the edge or at the point where arms were flexed or where legs or trunk were elevated, and this was the moment when maximum duration attained by the participant was recorded.

### Training

Test and training sessions were always performed at the same time of day, and the participants were told not to change their daily habits for the length of the study. These methods were not intended to substitute conventional climbing training but rather to serve as an additional resource for strength training. Afterwards, the other technical and physical training for that session was carried out.

Both methods of maximum finger strength training compared in this study shared apparatus, intensity and training volume: dead hanging exercises with the half crimp grip twice a week for 8 weeks with a regime of 3–5 sets, an effort duration of 10 s per set never reaching the point of muscular failure (leaving 3 s in reserve according to González-Badillo's line of research about the *perceived effort*; González-Badillo & Ribas, 2002), and a pause of 3 min between sets. Half crimp was chosen because it is the most widely used grip in climbing to grab small edges (Quaine & Vigouroux, 2004; Schweizer, 2001; Watts, 2004),



Figure 2. Execution of strength test (ST).



Figure 3. Depth used for the strength test: 15 mm.



Figure 4. Depth used for endurance test: 11 mm.

and is safer than the full crimp. The methods differed in the adjusting of the load: one used minimum edge depth without added weight (MED) and the other used bigger edge with maximum added weight (MAW). Group A ( $n = 5$ ) performed MED for 4 weeks (training phase 1), and for the following 4



Figure 5. Execution of endurance test (ET).

weeks (training phase 2: week 6–9) performed MAW (Group A referred to as MED–MAW). Meanwhile, Group B ( $n = 4$ ) used both methods in reverse order (Group B referred to as MAW–MED).

*Training with MAW.* It was performed on an 18-mm edge, after 3–4 warm-up sets of the training exercise with progressively heavier weight (50%–90% of the previous session's added weight). Each participant chose their added weight for the first training set of each session as follows: the weight should allow participants to hang for 13 s, but the effort duration would be for 10 s in any case; if this perceived 3-s margin was to be exceeded, the added weight for the next set would be increased (2–5 kg as a function of body weight). In contrast, if the margin were to be closer to 0 s, some weight would be removed (2–5 kg as function of body weight). As a reference, the strongest climber used up to 55 kg and the least strong used up to 25 kg. During the following sets that made up the session the weight was varied if needed so that the perceived effort was even throughout the training session. The number of sets per session is detailed in Table IV.

*Training on MED.* It was performed without additional weight. It was preceded by 3–4 warm-up sets of the training exercise, each with a progressively smallest edge (8–2 mm deeper than the one used in the previous session). Each participant chose the minimum depth for the first training set as follows: the first edge was chosen so that he or she could hold it for 13 s but would hang just for the prescribed 10 s. The depth of the edge for the next set could be increased or decreased by 1 or 2 mm according to the perceived effort. If the perceived 3-s margin was exceeded, the depth of the edge was decreased by 1–2 mm; if it approached zero, the depth was increased by 1–2 mm. The edge size was adjusted from one set to the next in order to control and maintain the perceived effort throughout the training session. The smallest edge that an individual used had a depth of 5 mm, and the biggest, 10 mm. The number of sets per session is detailed in Table IV.

*Technical and physical training.* It was carried out 6 days a week (Friday being rest day) 2–4 h a day, observing a 30-min rest period after the finger maximum strength training. It was performed on an artificial climbing wall and consisted of boulder, endurance and/or power endurance training with 3–10 sets of 3–90 moves with a difficulty between 70% and 100% of the maximum sport level. This training was tailored to the characteristics and goals of each participant, and was designed, revised and managed by Eva López.

### Statistical analysis

Descriptive statistics were used to derive means and standard deviation for all variables.

Reliability of ST and ET was estimated by using a one-way random repeated measures analysis of variance (ANOVA) and intra-class correlation coefficient for test–retest. To detect intra- and inter-group differences a repeated measure ANOVA with Bonferroni adjustment was used. The effect size between pre- and post-training for each group was calculated using Hedges'  $g$  (Hedges & Olkin, 1985), represented by the following formula:  $g = M_{\text{post}} - M_{\text{pre}} / \text{SD}_{\text{pooled}}$ , where  $M_{\text{post}}$  is the mean post-training measure,  $M_{\text{pre}}$  is the mean pre-training measure for each group and  $\text{SD}_{\text{pooled}}$  is the pooled SD of the pre- and post-measurements. The effect size was defined as: trivial,  $< 0.25$ ; small,  $0.25$ – $0.50$ ; moderate,  $0.50$ – $1$  and large,  $> 1$  according to the scale for strength training interventions on highly trained (at least 5 years of training) individuals proposed by Rhea (2004). Pearson's correlation was applied to investigate the relationships between variables. Differences between means and correlations were considered significant when  $p < 0.05$ .

### Results

Reliability of ST was estimated to be 0.93 (ICC) for one measurement (95% confidence interval 0.77–0.98) and 0.96 for the mean (95% confidence interval 0.87–0.99); the coefficient of variation (CV) was 7.8%. For ET, ICC for one measurement was 0.86 (95% confidence interval 0.56–0.96) and 0.92 for the mean (95% confidence interval 0.72–0.98); CV was 12.8%. No significant differences were found in strength and endurance of both intra- and inter-groups for the initial tests. Table V shows the results (mean  $\pm$  standard deviation) for the maximum strength and endurance tests.

The greatest maximum strength changes in percentage were obtained by MAW–MED group in ST2 and ST4 (9.6% and 6.9% against 2.1% and 0.6% for MED–MAW group). Both groups had strength losses in ST5, especially MED–MAW group (–5% against –0.3% for MAW–MED group). Although the MAW–MED group increased the most, no statistical differences were found in both intra- and inter-groups. Both groups demonstrated greater gains in maximum finger strength between ST1 and ST2 than between ST1 and ST3.

The two groups showed a remarkable improvement in endurance that did not result in statistically significant differences, both intra- and inter-groups. MAW–MED group demonstrated greater gains in ET2 and ET3 (16.69% and 19.95%, respectively), and MED–MAW group in ET3 (16.30%). Both lost endurance in ET5, particularly MAW–MED group (–2.59%).

The biggest losses compared to the maximum attained values occurred in Week 14, four weeks after

Table V. Results (mean  $\pm$  sd) for maximum strength test (kg) and endurance test (seconds).

|                  |     | MED–MAW group     |       | MAW–MED group     |       |
|------------------|-----|-------------------|-------|-------------------|-------|
| Maximum strength |     | kg                | ES    | kg                | ES    |
| Initial test     | ST1 | 39.68 $\pm$ 17.52 |       | 48.52 $\pm$ 11.31 |       |
| Training         | ST2 | 40.50 $\pm$ 15.15 | 0.1   | 53.18 $\pm$ 12.78 | 0.4   |
|                  | ST3 | 40.50 $\pm$ 15.15 | 0     | 49.18 $\pm$ 12.81 | 0.1   |
| Detraining       | ST4 | 39.90 $\pm$ 16.23 | 0     | 51.88 $\pm$ 10.70 | 0.2   |
|                  | ST5 | 37.70 $\pm$ 16.67 | – 0.2 | 48.40 $\pm$ 9.60  | – 0.1 |
| Endurance        |     | seconds           | ES    | seconds           | ES    |
| Initial test     | ET1 | 51.81 $\pm$ 10.76 |       | 46.62 $\pm$ 14.53 |       |
| Training         | ET2 | 57.78 $\pm$ 11.48 | 0.2   | 54.40 $\pm$ 16.21 | 0.5   |
|                  | ET3 | 60.56 $\pm$ 16.05 | 0.3   | 55.92 $\pm$ 13.86 | 0.7   |
| Detraining       | ET4 | 60.25 $\pm$ 14.71 | – 0.1 | 50.61 $\pm$ 15.09 | – 0.4 |
|                  | ET5 | 51.24 $\pm$ 11.28 | – 0.5 | 45.41 $\pm$ 13.38 | – 0.8 |

MED–MAW group = 4 weeks training with the minimum edge depth without added weight, and then 4 weeks training with maximum added weight on 18 mm edge; MAW–MED group = 4 weeks training with maximum added weight on 18 mm edge, and then 4 weeks training on the minimum edge depth without added weight. The improvements in strength and endurance were not statistically significant ( $p > 0.05$ ). ES = effect size. The ES was defined as: trivial,  $< 0.25$ ; small,  $0.25$ – $0.50$ ; moderate,  $0.50$ – $1$ ; and large,  $> 1$  according to the scale for strength training interventions on highly trained individuals proposed by Rhea (2004).

quitting the training object of the study, in maximum strength ( $-9.9\%$  for MED–MAW group and  $-6.9\%$  for MAW–MED group compared to the maximum obtained in ST2) as well as in endurance (Group A,  $-18\%$  and Group B,  $-22.54\%$  compared to the maximum obtained in ET3).

Effect size (ES) was calculated to be 0.1 and 0.4 for MED–MAW and MAW–MED in ST2, respectively. For ST4, only MAW–MED group showed effects ( $ES = 0.2$ ). For ET2, both groups showed an increase during the first 4 weeks, especially MAW–MED group ( $ES = 0.5$  against 0.2 for MED–MAW group). During the following 4 weeks (ET3) both groups improved, but again MAW–MED group registered the larger ES (0.7) against a small ES (0.3) for MED–MAW group. MAW–MED group experienced the biggest losses with an  $ES = -0.4$  in ET4 and  $ES = -0.8$  in ET5 against  $ES = -0.1$  and  $ES = -0.5$  for MED–MAW group (Table V).

A nearly significant positive correlation was found between the results of the initial tests ST1 and ET1 ( $r = 0.59$ ;  $p = 0.06$ ). Nonetheless, given that the control for body weight could influence the degree of correlation, partial correlations were applied controlling for body weight to those variables, and also to the changes in strength and endurance for all stages; the relation between ST1 and ET1 increased and results were statistically significant ( $r = 0.85$ ;  $p = 0.016$ ). Controlling for body weight had a similar effect on the relation between the changes in strength from ST1 to ST2 and the changes in endurance from ET1 to ET2 ( $r = 0.76$ ;  $p = 0.046$ ); between changes in strength from ST1 to ST3 and changes in endurance from ET1 to ET3 ( $r = -0.76$ ;  $p = 0.048$ ); and between the changes in strength from ST1 to ST4 and the changes in endurance from ET1 to ET4 ( $r = 0.84$ ;  $p = 0.018$ ).

## Discussion

To the best of our knowledge, this is the first study to compare different grip strength training methods using a climbing-specific device and exercise in elite rock climbers. From the results obtained from this study, the most important findings have been:

- The most effective training sequence to improve grip strength and endurance is to do the 10-s dead hanging with added weight and an edge depth of 18 mm and then proceed to dead hanging without added weight on the smallest edge depth that allows the participant to suspend themselves for 10-s intervals.
- A highly significant correlation between changes obtained in strength and changes obtained in endurance.

To interpret and contrast the results, it is important for us to note two things: (1) the participants were of a high level, at which it is harder to improve, and (2) we have not found works that analyse climbing and the effect of strength training with loads, so we have reviewed other studies that have analysed this in other sports or at a more general level.

### Improvements in strength

The biggest improvements occurred at 4 weeks of training (ST2) for the groups ( $+2.1\%$  and  $+9.6\%$  for MED–MAW and MAW–MED, respectively). This is in accordance with a study that found improvements within the first weeks among athletes of medium and high levels who obtained training with loads higher than 80% of 1 RM (one repetition maximum; Ebben, Kindler, Chirdon, Jenkins, Polichnowski, & Ng, 2004; Judge, Moreau & Burke, 2003; Kraemer,

1997; Rhea, Alvar, Burkett, & Ball, 2003). These studies suggest that improvements during the first weeks can be explained by neural adaptation.

At 8 weeks, there were hardly any improvements compared to the gains after 4 weeks, but there were losses. The MAW–MED group suffered the biggest loss (−8.3% against 0.5% for the MED–MAW group), perhaps due to the change in stimulus, i.e. training without weights. The MED–MAW group, trained with added weight in this second stage, did not increase in strength, in contrast with what was observed in MAW–MED after added weight training. This is thought to be caused by the MED method producing an increased level of fatigue from which the participant could not recover enough to be able to work afterwards with bigger loads.

As for the improvements within each group, the biggest, remarkable gain was obtained at ST2 for MAW–MED group that trained with weight, which was almost 10%. This increase is comparable to that of Rhea et al. (2003) after a 15-week training programme where gains of 9.8% in 1 RM were obtained. Nevertheless, they are smaller than those measured by Judge et al. (2003) among high-level throwers after a maximum strength training where volume and intensity were raised during 16 weeks; they found a significant increase of 15% in maximum isometric strength. This may be due to the fact that the athletes were not of a high level, so bigger performance gains were to be expected. This is not the case in this study, where the participants were of a high level; thus, a 9.6% increase in maximum strength can be considered significant.

Another notable result was that of MAW–MED group's increase in strength in Week 12, after 2 weeks of rest (+6.9% against +0.6% for MED–MAW group). We believe that this can be attributed to the effect of the rest itself; and is in accordance with Gibala, MacDougall & Sale, (1994), who registered at the 6th and 10th day of a decrease in load, there were increases in strength of 7% and 9%; which stresses the need for a rest period for it to have an effect on strength performance.

However, at Week 14 or 4 weeks of rest, both groups suffered important losses (−4.99% in MED–MAW group and −0.27% in MAW–MED group). These losses are in agreement with those obtained in some works after 10 days or 3 weeks of rest (Gibala et al. 1994; Hakkinen, Alen, Kallinen, Newton & Kraemer, 2000; Hortobagyi, Stevenson, Fraser, Johns & Israel, 1993). An explanation for this decrease in strength is given by Hakkinen, Komi & Alen (1985a), who assess the effects occurred during the detraining period and found a significant correlation between the loss in maximum strength and the reduction in maximum neural activation ( $p < 0.05$ ) in the leg extensors due to the break in training stimulus.

The most important part of this discussion is the significant increase in strength obtained by MAW–MED group, who were the group that trained first with weight and then without it. The explanation of this great effect could be that the use of loads while in dead hanging from a deeper edge provokes major muscular activation and recruitment of motor units, which in turn causes a bigger increase in grip strength than hanging off a smaller edge without added weight. This coincides with studies showing that there is higher muscular activation and recruitment of motor units when training with added weight (Hakkinen et al. 1985a, 1985b; Sale, 1988). In addition, the added weight dead hanging differs more from the usual climbing training than the small hold dead hanging. This may provide an extra stimulus and lead to higher adaptation processes.

#### *Improvements in endurance*

With endurance, it was also the MAW–MED group who improved the most (16.69% and 19.95% against 11.53% and 16.90% for MED–MAW group in ET2 and ET3, respectively). These improvements are in line with the improvements of 13% in the distance run in Hickson et al. (1988), or 13% of the time until fatigue caused by lactic acid in the study by Osteras et al. (2002) on participants who performed previously a 3-week strength training routine of three series of 5 RM. An explanation of this fact can be that by increasing maximum strength, body weight represents a smaller load during the exercise, so less motor units need to be activated for the same load and there is a potential for recruitment of bigger numbers of non-fatigued units. This delays the intervention of Type II fibres (Hickson et al. 1988), and the build-up of lactic acid, allowing an extension of the time until exhaustion (Marcinik et al., 1991).

The MAW–MED group, by training without added weight on a smaller edge depth after having made improvements with added weight on a deeper edge, kept improving endurance, probably because of the similarity of the exercise being performed to that of the test.

However, after 2 and 4 weeks of detraining, the major losses are found in the MAW–MED group (−11.40% and −22.54% against 0.60% and −8% in MED–MAW group), which is in accordance with a number of studies where the participants significantly lost what had been gained after abandoning the training with weight (Hakkinen et al. 1985a, 1985b; Hakkinen et al., 2000 & Kraemer et al. 2002).

From a practical point of view, the improvements in strength and endurance for both groups can be considered to be of note because, as has already been shown, maximum grip strength (MacLeod et al. 2007;

Watts, 2004; Watts et al. 2003) and grip endurance (Grant et al. 1996; Schöffel et al., 2005; Wall, Starek, Fleck & Byrnes, 2004) have a high relationship with climbing sport level. However, probably due to the low number of participants (MED–MAW group,  $n = 5$ ; MAW–MED group,  $n = 4$ ), these findings were not statistically significant. Since the MAW–MED group showed the biggest increase in both grip strength and endurance, the training combination they followed (MAW–MED) can be inferred to be more effective than the opposite one (MED–MAW).

#### *Correlation between strength and endurance*

Apart from these findings, a positive correlation was found between initial grip strength and grip endurance. However, as it is the case in the study by Guidi (1994,  $n = 13$ ), it was possibly due to the small sample size that the finding was not significant. A greater relationship between these two variables could be hidden by the influence of body weight, which was tested by performing a partial correlation ( $r = 0.85$ ;  $p = 0.016$ ). These results are in line with those of Janot, Mermier, Parker & Robergs (1999), who found a positive relationship of 0.78 between the maximum strength divided by body weight and performance level. Likewise, other authors have observed that the relationship between finger strength and body weight is more significant at a high performance level (Lehner & Heyters, 1998; Mermier, Janot, Parker & Swan 2000; Watts et al., 1993). Thus, our results suggest that maximum strength plays a relevant role in climbing, which is amplified if the effect of body weight at the time the participant is able to hold the edge is excluded.

Another important finding is the positive significant correlation that exists between the changes in grip strength and endurance at 4 and 8 weeks of training, and after 2 and 4 weeks of detraining, controlling for body weight in all cases. The results match those of Robinson, Stone, Johnson, Penland, Warren & Lewis, (1995), who obtained a significant correlation of 0.75 between the improvement in 1 RM and the sum of all the power peaks during a 15-pedal stroke series on a cycle ergometer. In no other study, has a relationship been quantified, although numerous authors have observed that after strength training of  $3 \times 5$  RM, the improvements in strength come accompanied by improvements in endurance (Hickson et al. 1988; Marcinik et al., 1991; Osteras et al. 2002).

However, Alricsson, Harms-Ringdahl, Linder, Larsson, & Werner (2004) did not find a correlation between the improvement in strength and endurance although they did obtain improvements of 10% in isometric strength and endurance after training the neck flexors and extensors using  $3 \times 10$  repetitions. According to the above study, it could be attributed to the individual differences between participants, so it

seems that the strongest participants were not the ones with the longest endurance time, and vice versa.

#### **Conclusions and practical applications**

The most effective method is to perform the training first on a bigger edge with added weight and then on a smaller edge without weight. The improvements were not statistically significant because of the small number of participants (MED–MAW group,  $n = 5$ ; MAW–MED group,  $n = 4$ ). Training maximum strength on a bigger edge improves endurance on smaller edges as well, as it is obvious from this fact that there exists highly significant correlation between changes in strength and changes in endurance.

To increase strength and endurance by 10% and 20% can be considered a remarkable improvement for high-level athletes, because both the ability to hold smaller edges and maintain grip strength for longer periods of time often determine a successful climb and make for substantial differences in competition. Also, it is probably not convenient to quit maximum strength training with weight for more than 2 weeks.

To the best of our knowledge, this is the first study to compare different grip strength training methods using a climbing-specific device in elite rock climbers. Our findings may help coaches to better prescribe finger strength training programmes in sport climbing. Another advantage is that many climbers can easily build this device and put in practice these methods.

#### **Future studies**

It would be interesting to check if there were significant inter- and intra-group differences if a larger number of participants trained for a longer time using the training method found as most effective in this research.

It would also be of interest to assess the effects of three different training methods on maximum finger strength and endurance over an 8-week cycle during which Group A would perform only the maximum strength training method suggested in this research as the most beneficial (MAW–MED), Group B would undergo endurance training with moderate loads and incomplete recovery and Group C would combine both types of training.

Judging the validity of the training presented in this study by checking its effects over performance on an actual climbing route would also be necessary, given that such a study is yet to be undertaken.

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